Fiscal Year:	FY 2021	**Task Last Updated:**	FY 01/18/2022
PI Name:	Lee, Michael Ph.D.		
Project Title:	Microbial Evolution and Transmission Aboard the ISS: Inferring Mutation Rates, Assessing Pangenomes, and Tracking Microbiome Transmission between Astronauts and the Space-based Built Environment		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mike.lee@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-4997
Organization Name:	Bay Area Environmental Research Institute/NASA Ames Research Center		
PI Address 1:	Mail Stop 239-4		
PI Address 2:	Bldg 239/ Room 379, P.O. Box 1		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-MOBE. Appendix B: Post-Doctoral Fellowships to Study the Microbiome of the ISS as a Built Environment
Start Date:	09/13/2018	**End Date:**	07/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Everroad, Craig	**Contact Phone:**	650-604-4997
Contact Email:	craig.everroad@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 6/15/20)		
Key Personnel Changes/Previous PI:	August 2019 report: There have been two personnel changes on the project as a whole. Dr. Alexander Voorhies was initially a co-investigator as he was heavily involved with the Astronaut Microbiome Project under Principal Investigator (PI) Dr. Hernan Lorenzi at the J. Craig Venter Institute (JCVI). Alexander has since moved on from JCVI and as such is no longer involved with this project. This hasn't hindered the work we are doing in integrating the astronaut microbiome data with our ISS isolate data. Another change is that I have closely partnered with Dr. Aubrie O'Rourke. Aubrie was awarded funding from the same MoBE call (hers through Sloan). She works out of JCVI in San Diego, CA where co-I Christopher Dupont is located, who connected us after we were both funded. Aubrie is very much "wet-lab" biologist, who was focusing on doing physiology and phenotypic assays working with Burkholderia isolated from the ISS, while I am very much a bioinformatician who was focusing on doing computational analyses with Staphylococcus and Ralstonia isolated from the ISS. We have since merged our isolate libraries, both working on all, and all of the science coming out of it is for the better. Neither of these personnel changes involves any budgetary changes. July 2020 report: No changes. July 2021 report: No changes.		

COI Name (Institution):	Everroad, Craig Ph.D. (MENTOR: Bay Area Environmental Research Institute/NASA Ames Research Center) Dupont, Christopher Ph.D. (J Craig Venter Institute, Inc.)
Grant/Contract No.:	80NSSC18M0110 (note funded under BAERI grant)
Performance Goal No.:	
Performance Goal Text:	
Task Description:	POSTDOCTORAL FELLOWSHIP NOTE: This project is funded by NASA as a result of a Space Act Agreement between NASA and the Sloan Foundation, for Collaborative Research Proposal for Microbiology of the Built Environment (MoBE). The purpose of this Space Act Agreement is to enhance the ability of NASA and the Sloan Foundation to support scientific research and the dissemination of information about the effect of spaceflight on microbial ecosystems in spacecraft such as the International Space Station (ISS) through sponsorship of postdoctoral fellowships. NASA is providing the isolates for all studies. OVERVIEW The proposed work will offer genetics-based avenues to gain insight into all 3 of the guiding questions as listed in this solicitation, but will directly address: “Does long-duration spaceflight alter normal rates of evolutionary change?” We propose to perform whole-genome sequencing of 50 isolates from the International Space Station (ISS) focused within the genera Staphylococcus and Ralstonia. These ISS-isolate genomes will be analyzed within the context of both an exhaustive, Earth-based reference genomic library constructed for each genus, and ISS astronaut microbiome metagenomic data. Co-Investigator Alexander Voorhies and collaborator Hernan Lorenzi are currently working with ‘before,’ ‘during,’ and ‘after’ microbiome metagenome from 9 ISS astronauts that were onboard the space station for various lengths of time spanning 2013–2016. Our unique position of being able to integrate microbiome sequencing data from another NASA effort with newly sequenced genomes of ISS isolates offers an unprecedented opportunity to explore many Space Biology evolutionary questions at the molecular level. OBJECTIVES/METHODS Genera Staphylococcus and Ralstonia were selected based on several criteria including ubiquity within human microbiota, potential pathogenicity, and their abundant representation within Earth-based reference databases, the ISS archive of available isolates, and our currently in-hand astronaut microbiome metagenomic data. We will investigate many metrics of molecular evolution aboard the ISS by directly addressing the following questions: 1) Are any particular genetic elements (i.e., genes/gene cassettes/operons) disproportionately represented in ISS isolates as compared to their Earth-based, nearest phylogenetic relatives? 2) Are there different propensities for specific individual genome mutations (i.e., single-point mutations such as substitutions/insertions/deletions, non-synonymous to synonymous substitution ratios, transition to transversion ratios, and genetic rearrangements) for organisms in the space-based built environment? 3) Which specific genes (and codons within these genes) are under positive, negative, or neutral selection aboard the ISS, but not within terrestrial references? 4) Can we track microbial transmission from an individual astronaut to the built environment, and/or the ISS serving as a vector and contributing to the microbiome of resident astronauts? 5) As assessed via comparison to metagenome-assembled genomes reconstructed from ISS-astronaut microbiomes, how representative are the isolate genomes recovered from the ISS of the actual microbiota of ISS astronauts?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	SIGNIFICANCE Microorganisms are not only integral to human health, but are also inexorable and necessary components of any self-contained elemental cycling, food production, or water system, and longer-duration human space exploration will require all of these systems. To design safe, built-environment microbial ecosystems to support these life-support systems requires not only understanding short-term, “plastic” microbial responses to the space environment, but also longer-term propensities for microbial evolution and adaptation. This work will leverage newly developed genetics-based computational tools, technologies, and approaches to generate the first comprehensive, whole-genomic database of microorganisms isolated from the space-based built environment while addressing fundamental questions about evolution aboard the ISS.
Task Progress:	Update: The primary work from this time period is described in our publication “Reference-guided metagenomics reveals genome-level evidence of potential microbial transmission from the ISS environment to an astronaut’s microbiome” (Lee et al. 2021). Using assembled genomes of our ISS-derived microbial isolates as references and recruiting metagenomic reads from an astronaut’s nasal microbiome revealed no recruitment to a Staphylococcus aureus isolate from samples before launch, yet systematic recruitment across the genome when sampled after 3 months aboard the ISS, with a median percent identity of 100%. This suggests that either a highly similar S. aureus population colonized the astronaut’s nasal microbiome while the astronaut was aboard the ISS or that it may have been below detection before spaceflight, instead supporting a shift in community composition. This work highlights the value in generating genomic libraries of microbes from built-environments such as the ISS and demonstrates one way such data can be integrated with metagenomics to facilitate the tracking and monitoring of astronaut microbiomes and health. Lee et al. (2021). Reference-guided metagenomics reveals genome-level evidence of potential microbial transmission from the ISS environment to an astronaut’s microbiome. iScience. https:// (see also Bibliography section below) We are still working on a publication focusing on the genomics and physiology of Staphylococcus isolates, similar to

	the publication focused on Burkholderia, previously reported in FY2020 report (O'Rourke A et al. "Genomic and phenotypic characterization of Burkholderia isolates from the potable water system of the International Space Station." PLoS One. 2020 Feb 20;15(2):e0227152. https://
Bibliography Type:	Description: (Last Updated: 01/19/2022)
Articles in Peer-reviewed Journals	Lee MD, O'Rourke A, Lorenzi H, Bebout BM, Dupont CL, Everroad RC. "Reference-guided metagenomics reveals genome-level evidence of potential microbial transmission from the ISS environment to an astronaut's microbiome." iScience. 2021 Feb 19;24(2):102114. https://doi.org/10.1016/j.isci.2021.102114 ; PMID: 33659879; PMCID: PMC7892915 , Feb-2021